Coherent adjacent-charge products in the memory core
The cross with the derivative singlet

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Abstract

We compute the previously missing equal-time cross between the output-rank deficit and the derivative singlet A. Adjacent-charge covariance composition gives a coherent Racah transform. Its signed first moment has an exact contiguous reduction, which preserves the cancellation of two terms of order m. At half filling the cross, after the energy-square subtraction, equals the previously obtained output-deficit variance. The next calculation is the norm of their difference in this same sector.

1 Operators and the result

Use standard CAR, the full overlapping complex Gaussian p-body ensemble, $\hbar = 1$, and the normalized maximally mixed trace in charge $m = N/2$. Inner products are $\langle U, V\rangle_m = \mathbb{E}\,\mathrm{tr}_m(U^\dagger V)$. Write $h = H_m$, $k = H_{m-1}$ for the *same* couplings, $C = \mathbb{E}h^2$, and

$$A = \mathrm{Rem}(k^2) + (2p - m)h^2, \qquad \mathrm{Rem}(X) = \sum_i c_i^\dagger X c_i,$$
$$O = \mathsf{W}_2 A, \qquad B = \mathsf{W}_2 h^2 = h^2 - CI. \tag{1}$$

Here W_2 denotes ordinary Gaussian second chaos, so $O = A - \mathbb{E}A$. For input ranks $a \le b$, let $Z^+_{ab;r} = P_r(h_a \odot h_b)$, where $\odot$ is a square for $a = b$ and an anticommutator otherwise. Set $S = a + b - r$ and $\mathcal{S}_N = \sum S\mathsf{W}_2 Z^+_{ab;r}$. Let $D_N = \sum_a (p - a)h_a$, $\bar{s} = \sum_a (p - a)C_a/C$, and $L_N = \mathsf{W}_2\{h, D_N - \bar{s}h\}$.

First take $p^2/N \to \lambda > 0$, with physical $C \to J_s^2/\lambda$; put $\theta = 2\lambda$, $q = e^{-4\lambda}$. We obtain

$$\begin{aligned} C^{-2}\langle B, O\rangle_m &\longrightarrow \theta(1 - q), \\ C^{-2}\langle L_N, O\rangle_m &\longrightarrow 0, \\ C^{-2}\langle \mathcal{S}_N, O\rangle_m &\longrightarrow \theta^2(1 + q) + \theta(1 - q). \end{aligned} \tag{2}$$

The final line is the output-deficit norm from the preceding note. The proof below computes the actual cross; it does not substitute an independent disorder copy for the two factors of A.

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2 A coherent covariance transform

Temporarily allow general m, and use independent, isotropic Hermitian rank-a and rank-b tensors U, V. Their restrictions $U^{\pm}, V^{\pm}$ to charges $m, m-1$ are coherently related primitive tensors, each normalized to $\mathbb{E}(U^{\pm})^2 = I$. Let $R = \min(m-1, N-m)$, $D_+ = \binom{N}{m}$, $D_- = \binom{N}{m-1}$, and define

$$d_r = \binom{N}{r}^2 - \binom{N}{r-1}^2, \qquad e_t = \binom{N}{t}\left[\binom{N}{t+1} - \binom{N}{t-1}\right],$$
$$\ell_r^2 = \frac{m(m-r)(N-m+1-r)}{N-m+1}. \tag{3}$$

The charged $(t+1, t)$ covariance eigenvalue of $\Psi_a(X) = \mathbb{E}U^+XU^-$ is $\zeta_a(t) = \ell_a T_a(t)/m$, where

$$T_a(t) = {}_4F_3\left(\begin{matrix}-a, a-N-1, -t, t-N \\ 1-m, m-N, -N-1\end{matrix}; 1\right). \tag{4}$$

One way to fix this normalization uses the known embedded-ensemble charged covariance, $\binom{m-t-1}{p}\binom{N-m+p-t}{p}$. If $\omega_a^{(p)}$ are the usual upper-sector rank weights, its triangular pure-rank decomposition reads

$$\sum_{a=0}^{p} \omega_a^{(p)} T_a(t) = \frac{m\binom{m-t-1}{p}\binom{N-m+p-t}{p}}{(m-p)\binom{m}{p}\binom{N-m+p}{p}}. \tag{5}$$

Substitution of the terminating series and interchange of the finite (a, j) sums gives this binomial identity (Vandermonde summation). The coefficient of T_p is nonzero, so the triangular system fixes all pure-rank eigenvalues. This uses established embedded-ensemble recoupling machinery, not a new ensemble prescription [1].

Define the coherent output contraction

$$M_{ab}(r) = \mathbb{E}\langle P_r(U^+V^+), \mathrm{Rem}\, P_r(U^-V^-)\rangle_m, \qquad \mu_{ab}(r) = \frac{mM_{ab}(r)}{\ell_a \ell_b}. \tag{6}$$

Composition of Ψ_a, Ψ_b, followed by covariance reshuffling, gives

$$T_a(t)T_b(t) = \sum_r \mu_{ab}(r) T_r(t),$$
$$\mu_{ab}(r) = \frac{d_r \ell_r^2}{m^2 D_+ D_-} \sum_{t=0}^{R} e_t T_a(t) T_b(t) T_r(t). \tag{7}$$

Indeed reshuffling is an HS isometry and $\sum_t e_t \zeta_r(t)\zeta_s(t) = D_+ D_- \delta_{rs}/d_r$. This proves the inversion without selecting tensor-product multiplicity bases or discarding their phases. We only use $\sum_r \mu_{ab}(r) = 1$; no positivity assumption is needed.

3 Exact unsigned and signed first moments

Return to $N = 2m$, $a, b < m$. Put $x_r = r(N + 1 - r)$, $X = x_{a+b} - x_r$, and let χ_{ab} be the previous neutral Racah kernel, with lower parameters $(-m, -m, -N - 1)$. Write

$$\xi_{ab} = {}_4F_3\left(\begin{matrix}-a, a-N-1, -b, b-N-1 \\ 1-m, -m, -N-1\end{matrix}; 1\right), \quad C_f = \frac{m^2(N+1)}{N}, \quad C_t = \frac{m(m-1)(N+1)}{N-1}. \tag{8}$$

With $\varepsilon = (-1)^{a+b-r}$, the four moments are

$$\begin{aligned} \sum_r \mu_{ab}(r) &= 1, & \sum_r X\mu_{ab}(r) &= -2ab + \frac{x_a x_b}{C_t}, \\ \sum_r \varepsilon\mu_{ab}(r) &= \xi_{ab}, & \sum_r \varepsilon X\mu_{ab}(r) &= -2ab\xi_{ab} + m(\xi_{ab} - \chi_{ab}). \end{aligned} \tag{9}$$

The unsigned identity follows by evaluating (7) at $t = 1$, since $T_a(1) = 1 - x_a/C_t$.

Here is a difference-equation derivation of the signed identities. The Racah equation [2], with $(\alpha, \beta, \gamma, \delta) = (-m, -m-1, -N-2, 0)$, yields

$$\begin{aligned} (\mathcal{L}f)_a &= B_a(f_a - f_{a+1}) + D_a(f_a - f_{a-1}), \\ B_a &= \frac{(a+1-m)(a-N-1)^2}{2(2a-N-1)}, & D_a &= \frac{a^2(a-m-2)}{2(2a-N-1)}, \end{aligned} \tag{10}$$

with $B_{m-1} = D_0 = 0$. Its eigenvalue on $T_a(t)$ is $y_t = t(N - t)$. In particular $2(B_a + D_a) = (m-1)(N+1) - x_a$. The alternating completeness sum, obtained by substituting the terminating series in (4), is

$$F(t) := \sum_r (-1)^r d_r \ell_r^2 T_r(t) = (-1)^{m-1+t} m D_-(m-t). \tag{11}$$

The same sum with x_r inserted is $[(m-1)(N+1) - y_t]F(t)$. Substitution in (7) gives the signed zeroth moment and the signed x_r moment $(x_a + \mathcal{L}_a)\xi_{ab}$.

For explicit reduction, the two charged components of a neutral rank-b tensor give

$$\xi_{ab} = T_a(b) - \frac{b}{N+1-2b}[T_a(b-1) - T_a(b)]. \tag{12}$$

Thus $(\mathcal{L}_a - x_b)\xi_{ab} = \frac{x_b}{N+1-2b}[T_a(b-1) - T_a(b)]$. If c_j is the jth summand of χ_{ab}, the corresponding summand of ξ_{ab} is $mc_j/(m-j)$. Termwise subtraction shows

$$\frac{x_b[T_a(b-1) - T_a(b)]}{N+1-2b} = -\sum_j \frac{jm}{m-j} c_j = -m(\xi_{ab} - \chi_{ab}), \tag{13}$$

which proves the last formula in (9), including $b = 0$.

4 Same-disorder cross and cancellation

In the actual p-body ensemble let $d = m - p > 0$, $C_a = \mathbb{E}h_a^2$, and $\omega_a = C_a/C$. Primitive embedding gives

$$\sqrt{\frac{C_a^-}{C_a}}\sqrt{\frac{C_b^-}{C_b}}\frac{\ell_a \ell_b}{m} = \frac{d^2}{m} =: g. \tag{14}$$

This independence of a, b is useful. The upper half-filled product obeys $P_r(V_b U_a) = \varepsilon P_r(U_a V_b)$. Consequently the same-$J$ cross, including the $a = b$ Gaussian decoupling coefficient, is

$$\Big\langle \sum f_{ab;r} \mathsf{W}_2 Z^+_{ab;r}, O \Big\rangle_m = C^2 \sum_{a,b} \omega_a \omega_b \sum_r f_{ab;r}(1+\varepsilon)[g\mu_{ab}(r) + (2p-m)\pi_{ab}(r)]. \tag{15}$$

Here π is the previous upper-sector ordered-product probability; pad μ by zero at $r = m$. The two terms in brackets must be combined before an asymptotic expansion.

Set $\eta = p^2/m$, $\Delta = \xi_{ab} - \chi_{ab}$, and $u_X = -2ab + x_a x_b / C_f$. The cross for $f = 1$ and $f = X/N$, per ordered input pair and divided by $C_a C_b$, is exactly

$$b_{ab} = \eta(1 + \chi_{ab}) + g\Delta, \tag{16}$$

$$z_{ab} = \frac{1}{N}\left\{\eta(u_X - 2ab\chi_{ab}) + g\left[\frac{x_a x_b}{m(m-1)(N+1)} + (m - 2ab)\Delta\right]\right\}. \tag{17}$$

We used $g + 2p - m = \eta$ and $C_t^{-1} - C_f^{-1} = [m(m-1)(N+1)]^{-1}$.

For fixed input deficits $p - a, p - b$ the terminating-series bounds give

$$\chi_{ab} \to q, \qquad m\Delta = \sum_j \frac{jm}{m-j} c_j \to -4\lambda q. \tag{18}$$

Each $|c_j|$ is bounded by $K_\lambda^j / j!$, uniformly for $0 \le a, b \le p$ at sufficiently large p. Also $m/(m-j)$ is uniformly bounded. Dominated summation therefore applies to the series and its first j moment. Equations (16)–(17) then imply

$$b_{ab} \to \theta(1-q), \qquad z_{ab} \to \theta^2(1+q) + \theta(1-q). \tag{19}$$

They also give uniform bounds $|b_{ab}| + |z_{ab}| \le K_\lambda$. The previous factorial input-deficit tail passes these limits through the full $\omega_a \omega_b$ sum. Multiplication by $2\bar{a} - a - b$, with $\bar{a} = \sum a\omega_a$, gives the L_N cross; its limit vanishes by the same weighted-tail bound.

5 From the Casimir proxy to the actual deficit

Define $\mathcal{X}_N = \sum X \mathsf{W}_2 Z^{+}_{ab;r}$. The finite numerical quantity in (17) is $\langle \mathcal{X}_N/N, O\rangle$. To obtain (2), use the exact identity

$$X = S[N + 1 - 2(a + b) + S]. \tag{20}$$

Orthogonality of input-pair and output-rank blocks, together with the preceding positive parity Gram, gives

$$\|\mathcal{S}_N - \mathcal{X}_N/N\|_{2,m} \le \frac{4p}{N}\|\mathcal{S}_N\|_{2,m} = O(N^{-1/2}). \tag{21}$$

We need only a weak bound on the other factor. The previous full-Fock, $\beta = \mu_{\mathrm{chem}} = 0$ calculation bounds $\|\mathsf{W}_2 A\|^2_{2,\mathrm{F}}$ at fixed positive λ. All sector contributions to this positive norm are nonnegative, so

$$\|O\|^2_{2,m} \le \frac{\|\mathsf{W}_2 A\|^2_{2,\mathrm{F}}}{2^{-N}\binom{N}{N/2}} = O(N^{1/2}). \tag{22}$$

Cauchy–Schwarz applied to (21) and (22) bounds the cross error by $O(N^{-1/4}) \to 0$. This step does not identify the full-Fock norm with a fixed-sector norm. The existing full-Fock result is used only as a positive bound.

6 What the cross tells us

Let $c = \theta(1 - q)/(1 + q)$, $\mathcal{S}_\perp = \mathcal{S}_N - cB$, and $Y = O - cB$. The preceding parity Gram and (2) yield

$$\begin{aligned} \lim\langle \mathcal{S}_\perp, Y\rangle_m &= \lim \|\mathcal{S}_\perp\|^2_{2,m} = V_S, \\ V_S &= J_s^4\left[\frac{2(1-q)}{\lambda} + \frac{16q}{1+q}\right] > 0. \end{aligned} \tag{23}$$

In particular $\langle \mathcal{S}_\perp, O - \mathcal{S}_N\rangle_m \to 0$. The output deficit is not orthogonal to the derivative singlet. The remaining scalar quantity for this part of the projection is

$$\Delta_N = \|O - \mathcal{S}_N\|^2_{2,m}. \tag{24}$$

If it has a finite limit Δ_*, the norm remaining after the B, O projection tends to $V_S\Delta_*/(V_S + \Delta_*)$. This is a conditional projection formula, not an evaluation of Δ_*. The already controlled input direction is orthogonal in this limiting Gram. Weighted A tails, the fixed-sector A norm, and the low-chaos terms of the raw memory coefficient remain to be computed. Only afterwards should the full energy-resolved spectrum and bulk mass dictionary be assembled.

7 Verification and continuation

All calculations here use the same CAR and covariance normalization as the earlier memory notes. The validation includes:

- 1,925 finite embedded/pure charged covariance identities at general charge, plus 139 orthogonality identities, 139 difference equations, and 556 signed/unsigned moments at half filling.
- Ten complete rectangular covariance maps, with 306,152 complex entries, checked using exact CAR and Casimir projectors. These test the map, not merely selected highest-weight vectors.
- The full Gaussian quadratic-coefficient Gram on $(N, p, m) = (4, 1, 2), (6, 1, 3), (6, 2, 3)$: respectively 136, 666, and 25,425 symmetric coordinate pairs. The B, $\mathcal{S}_N$, and $\mathcal{X}_N/N$ crosses with O agree with (15).
- 384 contiguous first-moment identities and two symbolic rational identities; three complete input-rank sums compared with an independent exact pair summation.
- Twelve complete-rank double-scale calculations at $\lambda = 0.1, 0.25, 0.5$ and $p = 16, 32, 64, 128$, evaluated at 80 and 120 decimal digits. Every rank and every terminating-series term is retained; the cancellation factor is recorded.

The last calculation uses the Casimir proxy. For example at $\lambda = 0.5, p = 128$, its cross divided by C^2 differs from its limit by about -0.0548285; the B and L_N cross deviations are about $1.95318\,10^{-5}$ and $3.32957\,10^{-5}$. These are finite-p deviations, not numerical errors. Precision refinement is not an interval-rounding certificate. The proxy is replaced by the actual deficit using the analytic bound on page 5.

The finite Gram gives positive residuals after projection onto B, O in all three test systems. This does not decide the double-scaled residual. The next calculation should evaluate Δ_N while preserving adjacent-charge interference, then control the g_A weighted tail from the previous core formula. No new time-dependent spectral measure or new bulk particle is claimed by this equal-time calculation.

References

[1] V. K. B. Kota and M. Vyas, *Random Matrix Theory for Transition Strength Densities in Finite Quantum Systems: Results from Embedded Unitary Ensembles*, arXiv:1504.00759, Secs. 3 and 7. The recoupling structure is used; its dilute limit is not our half-filled double scaling.

[2] R. Koekoek and R. F. Swarttouw, *The Askey-scheme of hypergeometric orthogonal polynomials and its q-analogue*, report 98-17 (1998), Sec. 1.2, Eq. (1.2.5).